kcat-Km Prediction for Unseen Enzymes: Solution Report

Summary

The nominated model predicts log10 kcat and log10 Km with a cross-validated mean Spearman of **0.588** (kcat 0.485, Km 0.692) on enzymes unseen in training, up from 0.395 for the starter ridge model. On the public leaderboard it scored **0.583** (sub-17).

It blends two models that run inside the grader's limits of 2 CPUs and 10 minutes: a fast model on ESM-2 8M protein embeddings and a stronger model on ESM-2 35M embeddings cut after layer 6. Both use ChemBERTa substrate embeddings, RDKit descriptors and label-derived neighbour features, and both train on extra public data whose labels were checked against the competition's.

Problem and evaluation

Each example is an enzyme's amino-acid sequence and a substrate's SMILES string; the targets are log10 kcat (turnover number, 1/s) and log10 Km (Michaelis constant, mM). The leaderboard score is the mean of the two Spearman correlations, so only the ranking of predictions matters.

Set	Rows	Enzymes	Labels
Training data	42,524	17,432	kcat and Km
Public validation inputs	1,901	475	none
Hidden test set	1,714	848	none (graded)

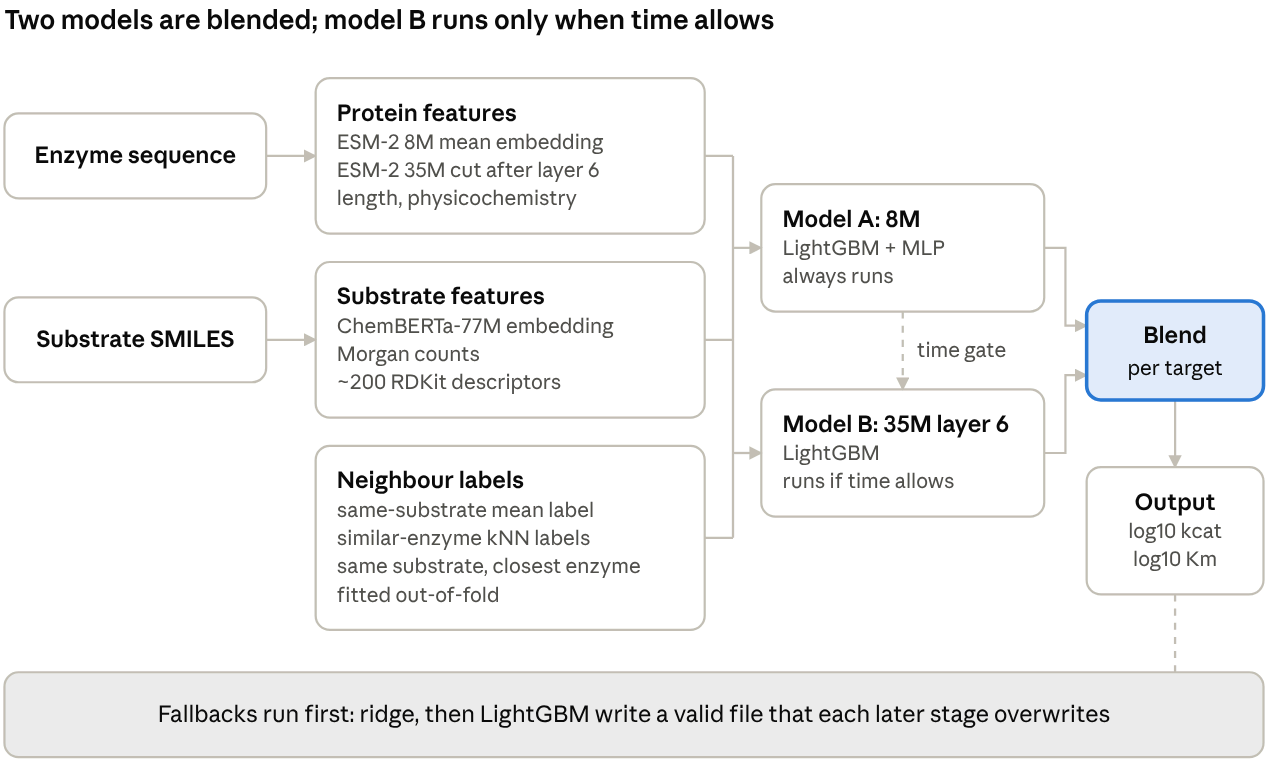
The test enzymes do not occur in training, so all local evaluation mimics that:

- **Split:** 5-fold GroupKFold over enzyme clusters. Two sequences are linked when their 3-mer cosine similarity is at least 0.3; connected components form 4,258 groups, and a whole group sits in one fold.
- **Comparisons:** every run uses the same cached folds, so two models are compared fold by fold. An idea is kept only when it wins on most folds (usually 5 of 5) by more than about 0.003.

- **Test-like score:** a second CV score that caps each enzyme at 2 rows, like the test set. It ranked runs almost identically to plain CV (Spearman 0.99 across runs), so plain CV was used.
- **Grader constraints:** the submission is a Hugging Face model repository whose `predict.py` the grader runs with 2 CPUs, 16 GB, no network and a 10-minute limit. Every build is tested in Docker under the same limits before upload.

Final solution

The submission computes features once per unique enzyme and substrate, runs a fast model and, when time allows, a stronger one, and blends them per target.



final submission · inference pipeline

- **Model A** (ESM-2 8M features): LightGBM with Huber loss and a multi-task MLP, averaged with per-target weights (kcat 0.75/0.25, Km 0.70/0.30). Trained on the competition data plus the extra rows.
- **Model B** (ESM-2 35M cut after layer 6): LightGBM with Huber loss, 1,500 trees, learning rate 0.03, 63 leaves, 30% of features per tree. Trained the same way.

- **Blend:** per-target weights tuned on out-of-fold predictions; model B carries 0.55 for both targets.
- **Time gate:** `predict.py` measures how long model A's protein embeddings took and runs model B only if its estimate (2.9 times that, from the two models' sizes plus a 30% margin) fits before the 540-second internal deadline. In the Docker dry run both models finished in 233 seconds.
- **Fail-safe:** a ridge model and a LightGBM model on hand-made features write a valid file first; each model runs in a child process with a timeout, so a crash or hang cannot leave the submission without output.

Development: five ideas that moved the score

Each idea below was kept only after it beat the previous best fold by fold on the fixed enzyme-cluster CV.

1. **Leakage-aware baselines.** The starter ridge model on composition features scored CV 0.395. LightGBM on hand-made protein features (length, amino-acid composition, physicochemistry) and substrate features (Morgan counts, about 200 RDKit descriptors) reached 0.499. This model is still bundled as a fallback.
2. **Pretrained embeddings and neighbour labels.** Mean-pooled ESM-2 protein embeddings and ChemBERTa-77M substrate embeddings were added. The largest single gain came from label-derived features, computed out-of-fold:
   - the smoothed mean label of the same substrate;
   - a similarity-weighted mean of the labels of the 10 most similar training enzymes;
   - among rows with the same substrate, the labels of the closest enzyme. LightGBM with these reached 0.562 (35M) and an 8M ensemble of LightGBM and a multi-task MLP 0.564.
3. **Two models within a time budget.** The 35M model was better but too slow to run alone on the grader's CPUs. The submission therefore runs the cheap 8M ensemble first and the 35M model only when the measured time leaves room, then blends them with per-target weights tuned on out-of-fold predictions (CV 0.577). Staged, fail-safe inference guarantees a valid file even if a stage crashes or times out.
4. **Middle-layer protein embeddings and model diversity.** Cutting ESM-2 35M after layer 8 (0.568) and then layer 6 (0.572) beat using all 12 layers (0.562), and was faster. ExtraTrees and CatBoost added diversity to the ensembles; the blend reached CV 0.582.

5. **Checked external data.** New pairs from public kinetics collections were added for training only, each source used only for the target whose labels agree with the competition's. CatPred Km raised Km by 0.016; with it in both models the blend reached CV 0.588.

Extra data

The final model trains on 22,786 extra rows from two public collections, each used only for the target whose labels match the competition's.

Source	Target	New pairs	Spearman vs competition labels	Used
CatPred	Km (mM)	15,105	0.989	yes
DLKcat	kcat (1/s)	8,514	0.955	yes
CatPred	kcat (1/s)	5,672	0.858	no: below the 0.9 bar
Leak-CURBER	Km	20,581	0.755	no

How the rows were prepared and used:

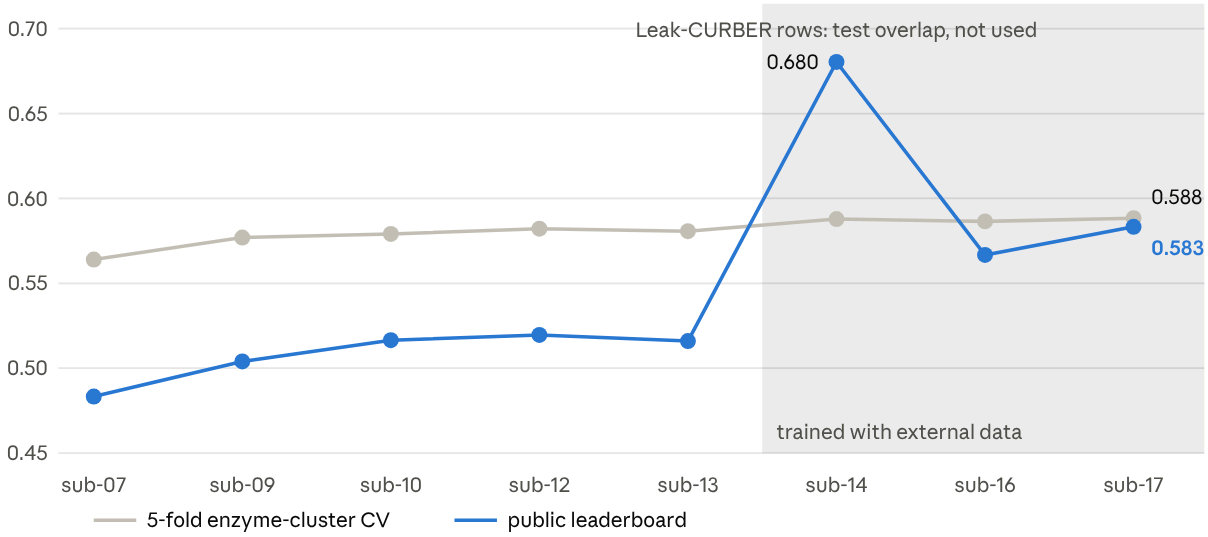
- **Matching:** pairs are compared on RDKit canonical SMILES plus the exact sequence, because collections spell the same molecule differently. An extra pair is kept only if it is in none of the competition training data, the Leak-CURBER benchmark files or the validation inputs.
- **Labels:** repeated measurements follow the competition's rules (kcat = maximum, Km = mean of raw values, then log10). A row with only one trusted label trains that target's models and is skipped by the other's.
- **Held-out pairs removed:** the 649 pairs that also appear in the validation inputs were dropped, so no extra pair coincides with a validation pair.
- **Honest CV:** extra rows are never scored, and an extra enzyme similar to a fold's validation enzymes is left out of that fold's training.

Results

The nominated blend reaches CV 0.588 (kcat 0.485, Km 0.692) and leaderboard 0.583; without external data the leaderboard sat 0.06 to 0.08 below CV every time.

CV and leaderboard rose together until external data lifted the leaderboard

Mean Spearman of each submitted blend, in submission order



run log and submission ledger · 8 submissions, 28 Oct to 1 Nov 2026

Up to sub-13 the gap was steady, so CV ranked models reliably. With external data the gap narrows or reverses, because public kinetics collections share pairs and enzymes with the held-out data; for those models only the CV gain is trustworthy.

Ideas that were measured and dropped:

Limitations and conclusions

A trustworthy, enzyme-grouped CV made many small gains reliable: each kept idea added 0.002 to 0.07, and together they took mean Spearman from 0.395 to 0.588.

- **kcat stays hard.** Its CV Spearman is 0.485 against 0.692 for Km. kcat depends on the reaction mechanism and assay conditions, which neither embedding captures.
- **Test enzymes are harder than the CV folds.** Without external data, the leaderboard sat 0.06 to 0.08 below CV, consistently, so CV ranked models correctly but overstates the absolute score.
- **Compute budget.** On 2 CPUs in 10 minutes, ESM-2 150M took about 1,600 s for the validation inputs alone, so larger protein models were out of reach.

- **Label noise.** Public collections disagree on shared pairs (DLKcat and CatPred kcat: Spearman 0.70), so extra data adds noise as well as signal.

Reproducibility

Every run is defined by a YAML config and logged with its config fingerprint, data fingerprint and per-fold scores, so any result can be rebuilt.

Part	Where	Role
Features	`src/kcatkm/features/`	protein, substrate and language-model extractors, cached per unique string
Neighbour labels	`src/kcatkm/labelfeat.py`	out-of-fold label-derived features
Models	`src/kcatkm/models.py`, `nn.py`	LightGBM, ExtraTrees, CatBoost, MLP, weighted ensembles
CV	`src/kcatkm/splits.py`, `experiment.py`	enzyme-cluster folds, runs, logging
Extra data	`src/kcatkm/extra.py`, `scripts/source_overlap.py`	import, label checks, training-only rows
Inference	`submission_template/predict.py`	staged, fail-safe, time-gated prediction
Packaging	`scripts/build_submission.py`	refit, local and Docker checks, upload

Rebuilding the nominated submission:

```
python scripts/source_overlap.py --download --write --public-only --exclude-
val --leakcurber <folder with the Leak-CURBER _1d files>
python scripts/run.py configs/ens_lf_esm8_pub.yaml
configs/lgbm_lf_l6_pub.yaml --no-val
python scripts/tune_weights.py ens_lf_esm8_pub
python scripts/build_submission.py ens_lf_esm8_pub_tuned --second
lgbm_lf_l6_pub --compiled --docker
```

The deliverables (readable code, the model weights and run records) are collected with `python scripts/make_deliverables.py <submission> --zip`.